End-to-End Clinical Reporting Pipeline: Raw Data → SDTM → ADaM → TLF

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Table of contents

1 Work Plan **3**
1.1 Phase 1: Alignment & Infrastructure (Week 1) . . . 3
1.2 Phase 2: SDTM Generation with `sdtm.oak` (Week 2) . . . 3
1.3 Phase 3: ADaM Creation with `admiral` (Weeks 3–4) . . . 4
1.4 Phase 4: TLF Dashboard with `teal` (Weeks 5–6) . . . 4
1.5 Phase 5: Integration & End-to-End Testing (Week 7) . . . 5
1.6 Phase 6: Final Delivery & Handover (Week 8) . . . 5
1.7 Communication & Meetings . . . 6
1.8 Success Criteria . . . 6

2 Complete R Code **6**

3 Dashboard Snapshorts **10**
3.1 DM Table . . . 10
3.2 AE Table . . . 11
3.3 Kaplan-Meier Plot . . . 11

1 Work Plan

Parties: Fuxi (pharmaverse/admiral/teal experience) & Thomas (R-Shiny/teal application)
Start Date: Immediately after April's 3rd week, 2026 meeting
Objective: Create a reproducible, interactive pipeline using `sdtm.oak` → `admiral` → `teal`, focusing on the DM/VS domain, ADSL, and a TLF dashboard.

1.1 Phase 1: Alignment & Infrastructure (Week 1)

Task	Owner	Deliverable
1.1 Review corrected `SDTM-Datasets-Creation-with-SDTM.OAK.pdf`	Both	Shared understanding of DM creation logic
1.2 Review `ADaM-Datasets-Creation-with-Admiral.pdf` (ADSL)	Both	Agreed ADSL derivation steps (exposure, disposition, death, flags, codelists)
1.3 Review `TLF-R-Shiny-Dashboard-with-Teal.pdf`	Both	Confirm dashboard modules: DM table, AE table, KM plot
1.4 Set up shared environment (Git repo, R package versions, data access)	Thomas	GitHub repo + `renv` lock file
1.5 Identify raw data sources (CDISC SDTM test data, e.g., `pharmaversesdtm`)	Fuxi	Documented input datasets

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1.2 Phase 2: SDTM Generation with `sdtm.oak` (Week 2)

Task	Owner	Deliverable
2.1 Run the corrected DM domain script on sample raw data	Fuxi	Validated DM dataset (SDTM)
2.2 Extend to other core domains (e.g., AE, EX, DS) if needed for ADaM	Fuxi	At least AE, EX, DS in SDTM format
2.3 Write QC checks (e.g., no duplicate USUBJID, expected variables)	Fuxi	QC script

Task	Owner	Deliverable
2.4 Document any deviations from the corrected script	Fuxi	Markdown log

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1.3 Phase 3: ADaM Creation with `admiral` (Weeks 3–4)

Task	Owner	Deliverable
3.1 Re-create ADSL following the PDF steps (exposure, disposition, death, flags, codelists)	Fuxi	ADSL dataset (one row per subject)
3.2 Add derived variables: `RACEGR1`, `AGEGR1`, `REGION1`, `DTHCGR1`, numeric codes	Fuxi	Completed ADSL with all planned variables
3.3 Generate ADSL metadata (xportr labels, formats) and write `.xpt`	Fuxi	`adsl.xpt` + metadata log
3.4 (Optional) Create additional ADaM datasets (ADAE, ADTTE) for the dashboard	Both	`adae`, `adtte` (if not already present)

1.4 Phase 4: TLF Dashboard with `teal` (Weeks 5–6)

Task	Owner	Deliverable
4.1 Adapt the provided `teal` code to use real ADSL/ADAE/ADTTE	Thomas	Running `teal` app locally
4.2 Configure pre-defined filters (`teal_slices`) as in PDF: `SAFFL`, `AESER`, `AERE1`, age, sex	Thomas	Filter panel with anchored slices

Task	Owner	Deliverable
4.3 Customize modules: demographic table, AE table, KM plot (add strata/facets)	Thomas	Fully functional dashboard
4.4 Add a module to display SDTM/ADaM metadata (variable labels, source)	Thomas	"Data Dictionary" tab

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1.5 Phase 5: Integration & End-to-End Testing (Week 7)

Task	Owner	Deliverable
5.1 Chain the pipeline: raw data → SDTM → ADaM → teal dashboard	Both	Single R script or `targets` workflow
5.2 Test with a second raw dataset (e.g., a different study)	Both	Validation report
5.3 Performance check (loading speed, filter responsiveness)	Thomas	Benchmark summary
5.4 Document known limitations (e.g., date imputation rules, missing value handling)	Fuxi	`README` + code comments

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1.6 Phase 6: Final Delivery & Handover (Week 8)

Task	Owner	Deliverable
6.1 Prepare a demo for stakeholders (recorded video or live)	Both	15-min demo
6.2 Package pipeline as an R package or Shiny app bundle	Thomas	Deployable artifact (e.g., Docker image)

Task	Owner	Deliverable
6.3 Write user guide for analysts (how to add new study data)	Fuxi	PDF or RMarkdown guide
6.4 Schedule follow-up for next features (e.g., additional domains, safety tables)	Both	Roadmap document

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1.7 Communication & Meetings

- **Weekly sync:** Every Friday, 1:30 PM PDT (30 min) – progress review, blockers, next steps.
- **Ad-hoc:** As needed via email or shared chat.
- **File sharing:** GitHub repository + shared drive for raw data (if not public).

1.8 Success Criteria

1. Raw data (SDTM-like) → DM domain (SDTM) runs without error.
2. ADSL is created with all variables shown in the PDF, and passes basic CDISC conformance.
3. `teal` dashboard launches and displays the three modules with pre-set filters.
4. A user can select a different raw dataset and re-run the entire pipeline with minimal changes.

2 Complete R Code

```r
# Load required package
library(teal.modules.clinical)

ADSL <- teal.modules.clinical::tmc_ex_adsl
ADAE <- teal.modules.clinical::tmc_ex_adae
ADTTE <- teal.modules.clinical::tmc_ex_adtte

app <- init(
  data = cdisc_data(
    ADSL = ADSL, ADAE = ADAE, ADTTE = ADTTE,
    code = "
      ADSL <- teal.modules.clinical::tmc_ex_adsl
      ADAE <- teal.modules.clinical::tmc_ex_adae
      ADTTE <- teal.modules.clinical::tmc_ex_adtte
    "
  ),
```

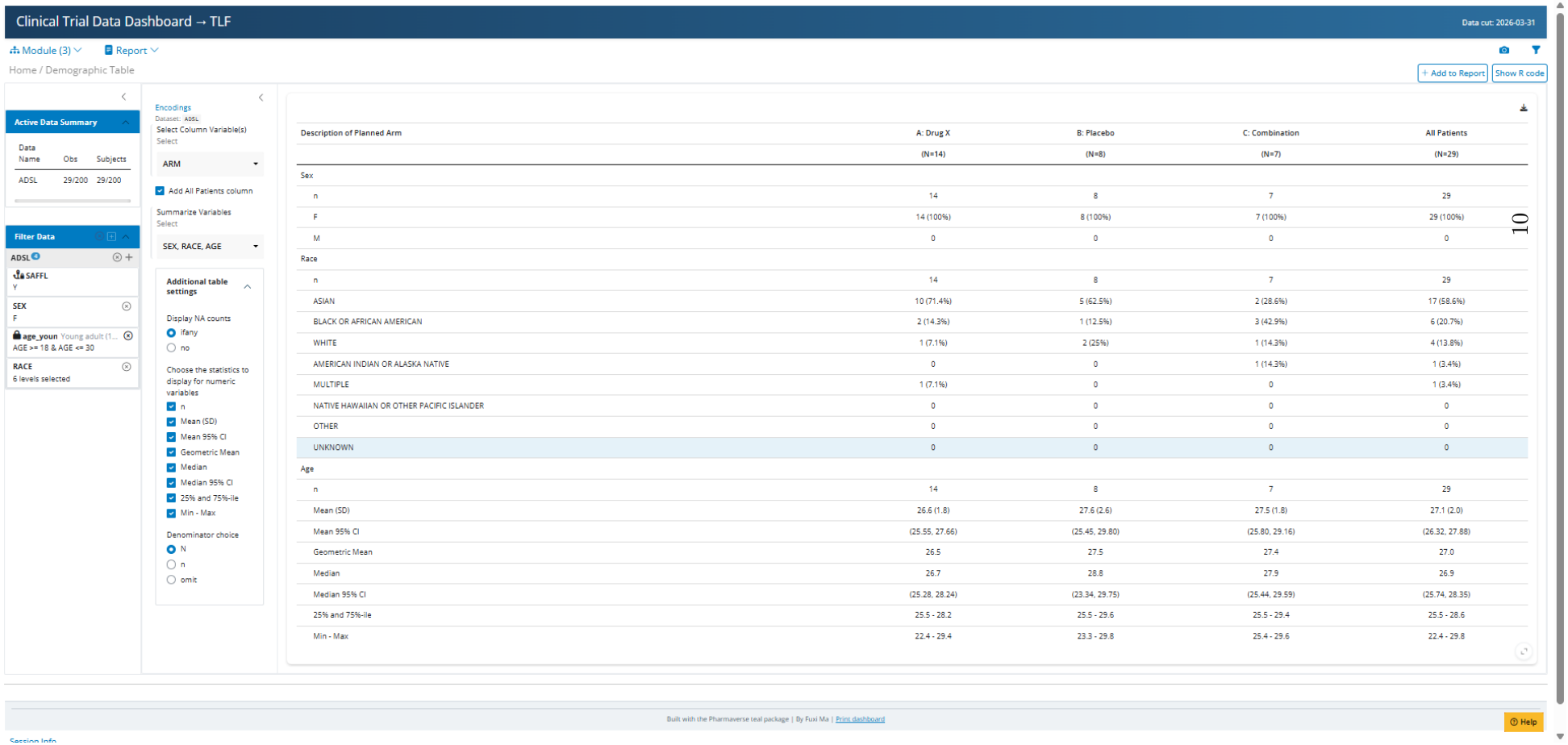
```r
modules = modules(
  tm_t_summary(
    label = "Demographic Table",
    dataname = "ADSL",
    arm_var = choices_selected(
      choices = c("ARM", "ARMCD"), selected = "ARM"
    ),
    summarize_vars = choices_selected(
      choices = c("SEX", "RACE", "BMRKR2", "EOSDY", "DCSREAS", "AGE"),
      selected = c("SEX", "RACE")
    )
  ),
  tm_t_events(
    label = "AE Table",
    dataname = "ADAE",
    arm_var = choices_selected(
      choices = c("ARM", "ARMCD"), selected = "ARM"
    ),
    hlt = choices_selected(
      choices = variable_choices(ADAE, c("AEBODSYS", "AESOC")),
      selected = "AEBODSYS"
    ),
    llt = choices_selected(
      choices = variable_choices(ADAE, c("AETERM", "AEDECOD")),
      selected = "AEDECOD"
    )
  ),
  tm_g_km(
    label = "KM Plot",
    dataname = "ADTTE",
    arm_var = choices_selected(
      choices = c("ARM", "ARMCD", "ACTARMCD"),
      selected = "ARM"
    ),
    paramcd = choices_selected(
      choices = value_choices(ADTTE, "PARAMCD", "PARAM"),
      selected = "OS"
    ),
    strata_var = choices_selected(
      variable_choices(ADSL, c("SEX", "BMRKR2")), NULL
    ),
    facet_var = choices_selected(
```

```r
        variable_choices(ADSL, c("SEX", "BMRKR2")), NULL
      ),
      plot_height = c(600L, 400L, 5000L)
    )
  ),
  header = tags$h1("Clinical Trial R-Shiny App"),
  footer = tags$div(
    style = "text-align: center; margin-top: 20px;
             padding: 10px; background-color: #f5f5f5;
             font-size: 12px;",
    "Fuxi Ma", "Built with Pharmaverse teal package"
  ),
  filter = teal_slices(
    teal_slice("ADSL", "SAFFL", id = "saffl",
               selected = "Y", fixed = TRUE, anchored = TRUE),
    teal_slice("ADAE", "AESER", anchored = TRUE),
    teal_slice("ADAE", id = "aerel",
               expr = "AEREL == 'Y' & AETOXGR %in% c('3', '4', '5')",
               title = "Grade 3+ Related Events"),
    teal_slice("ADSL", "SEX", id = "sex", fixed = TRUE),
    teal_slice("ADSL", id = "age",
               expr = "AGE >= 18 & AGE <= 30", title = "Young Adult"),
    module_specific = TRUE,
    mapping = list(
      "Demographic Table" = c("saffl"),
      "AE Table" = c("saffl", "aerel"),
      global_filters = c("sex", "age")
    ),
    count_type = "all"
  )
)

shinyApp(app$ui, app$server)
```

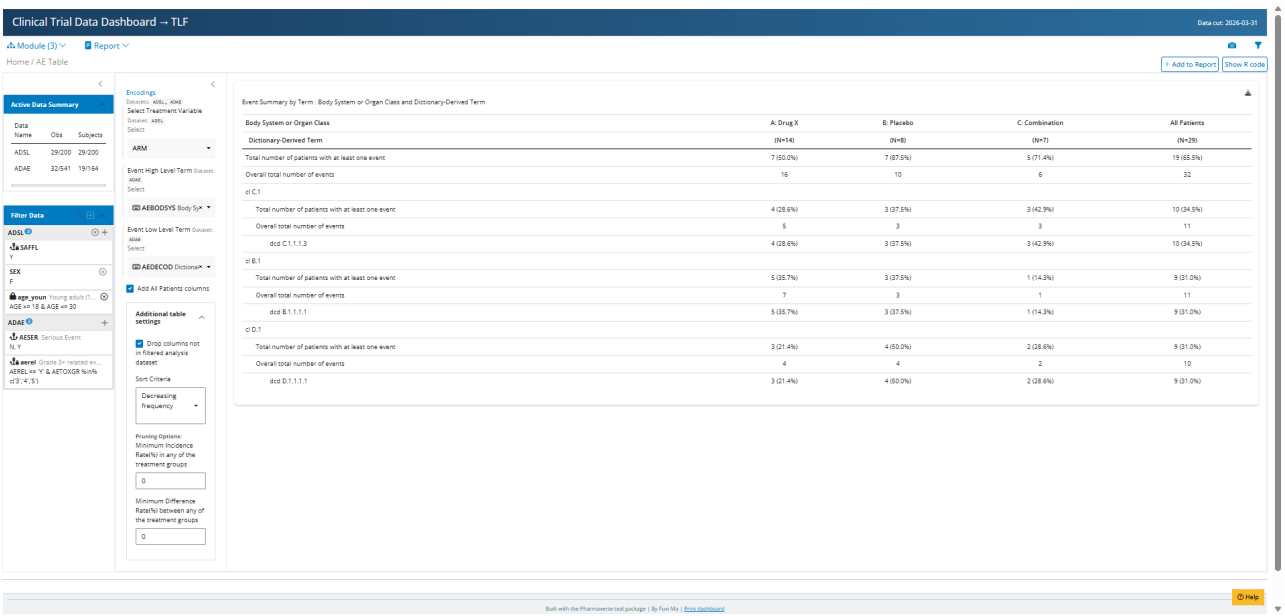
3 Dashboard Snapshots

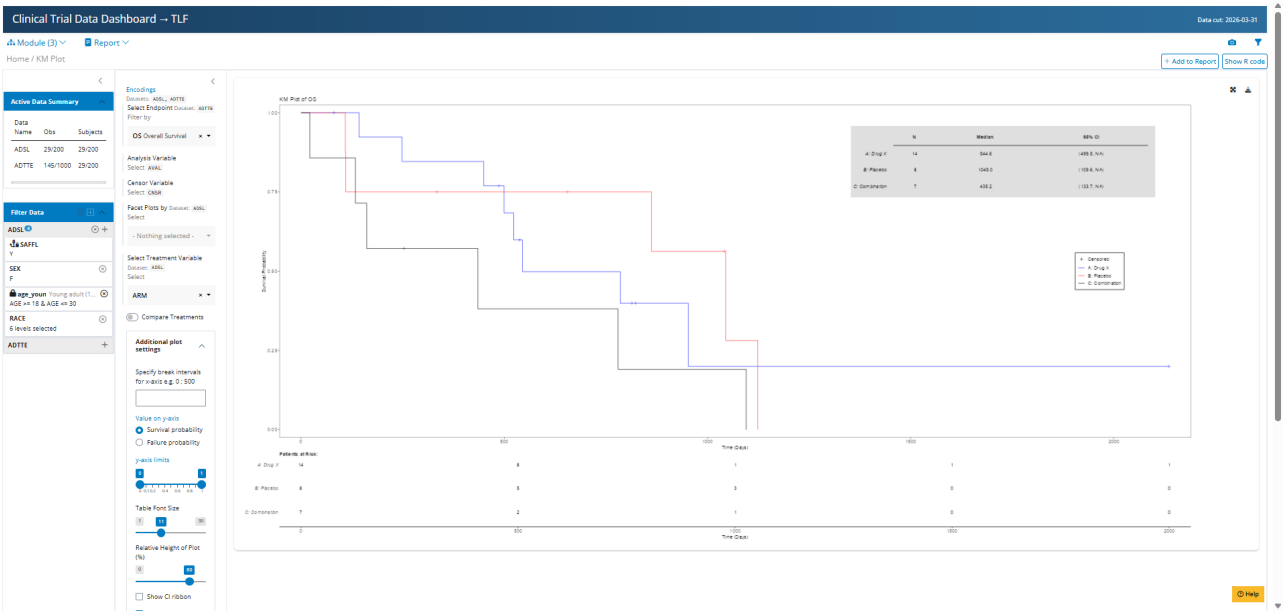
3.1 DM Table



Description of Planned Arm	A: Drug X	B: Placebo	C: Combination	All Patients
	(N=14)	(N=8)	(N=7)	(N=29)
Sex				
n	14	8	7	29
F	14 (100%)	8 (100%)	7 (100%)	29 (100%)
M	0	0	0	0
Race				
n	14	8	7	29
ASIAN	10 (71.4%)	5 (62.5%)	2 (28.6%)	17 (58.6%)
BLACK OR AFRICAN AMERICAN	2 (14.3%)	1 (12.5%)	3 (42.9%)	6 (20.7%)
WHITE	1 (7.1%)	2 (25%)	1 (14.3%)	4 (13.8%)
AMERICAN INDIAN OR ALASKA NATIVE	0	0	1 (14.3%)	1 (3.4%)
MULTIPLE	1 (7.1%)	0	0	1 (3.4%)
NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER	0	0	0	0
OTHER	0	0	0	0
UNKNOWN	0	0	0	0
Age				
n	14	8	7	29
Mean (SD)	26.6 (1.8)	27.6 (2.6)	27.5 (1.8)	27.1 (2.0)
Mean 95% CI	(25.55, 27.66)	(25.45, 29.80)	(25.80, 29.16)	(26.32, 27.88)
Geometric Mean	26.5	27.5	27.4	27.0
Median	26.7	28.8	27.9	26.9
Median 95% CI	(25.28, 28.24)	(23.34, 29.75)	(25.44, 29.59)	(25.74, 28.35)
25% and 75%-ile	25.5 - 28.2	25.5 - 29.6	25.5 - 29.4	25.5 - 28.6
Min - Max	22.4 - 29.4	23.3 - 29.8	25.4 - 29.6	22.4 - 29.8

3.2 AE Table



Body System or Organ Class	A: Drug X	B: Placebo	C: Combination	All Patients
Dictionary-Derived Term	(N=14)	(N=8)	(N=7)	(N=29)
Total number of patients with at least one event	7 (50.0%)	7 (87.5%)	5 (71.4%)	19 (65.5%)
Overall total number of events	16	10	6	32
cl C.1				
Total number of patients with at least one event	4 (28.6%)	3 (37.5%)	3 (42.9%)	10 (34.5%)
Overall total number of events	5	3	3	11
dcd C.1.1.1.3	4 (28.6%)	3 (37.5%)	3 (42.9%)	10 (34.5%)
cl B.1				
Total number of patients with at least one event	5 (35.7%)	3 (37.5%)	1 (14.3%)	9 (31.0%)
Overall total number of events	7	3	1	11
dcd B.1.1.1.1	5 (35.7%)	3 (37.5%)	1 (14.3%)	9 (31.0%)
cl D.1				
Total number of patients with at least one event	3 (21.4%)	4 (50.0%)	2 (28.6%)	9 (31.0%)
Overall total number of events	4	4	2	10
dcd D.1.1.1.1	3 (21.4%)	4 (50.0%)	2 (28.6%)	9 (31.0%)

3.3 Kaplan-Meier Plot